

Unit-1 Sample questions

1. Find the general equation of SHM. A force of 500N is used to stretch a spring with a 0.5 kg mass attached to it by 0.35m. (a) Calculate the time period of an oscillator (b) Calculate the frequency of the oscillator.
2. the (a) Amplitude, (b) Frequency, (c) Time period and (d) Spring constant.
3. Justify the statement "All SHM are oscillatory motions, however, all oscillatory motions are not SHM". A horizontal spring ($k = 300 \text{ N/m}$) with a mass of 0.75 kg attached to it is undergoing simple harmonic motion. Calculate the (a) Time period, (b) Frequency and (c) Angular frequency of this oscillator.
4. Describe the equation of forced vibrations. b) Describe the amplitude & phase of forced vibrations.
5. An oscillator is subjected to an external periodic force and a damping force proportional to the velocity. Set up an differential equation of the motion of the oscillator. Mention the conditions under which velocity resonance occurs.
6. Define forced oscillations. List the equations of motion associated with forced oscillations. Explain the concept of resonance and its impact on the amplitude of an oscillator.
7. What are damped oscillations? Derive the equation of motion and solution of the damped oscillator.
8. Graphically show the displacement-time curve for oscillatory motion, critically damped, and over-damped motion of a damped oscillator. Mention the conditions of their occurrence.
9. Graphically show the displacement-time curve for critically damped, and over-damped motion of a damped oscillator. A damped harmonic oscillator has a frequency of 5 oscillations per second. The amplitude drops to half its value for every 10 oscillations. Find the damping coefficient.
10. Find the difference between periodic motion, SHM and oscillatory motion with three examples each?
11. Derive the general equation and solution of SHM?
12. Obtain the mathematical equations for the potential energy, kinetic energy, and total mechanical energy of the system at any point in time for spring-mass system.
13. A spring-mass system has a spring constant of 200 N/m and a mass of 4 kg. The system is displaced by 0.1 m from its equilibrium position and released. Calculate the following:
 - (i) The natural frequency of the system.
 - (ii) The maximum potential energy stored in the spring.
 - (iii) The maximum kinetic energy of the mass during oscillation.
14. Give the graphical comparison among the following three types of damped harmonic motion and write the expression of the amplitude of motion for each of these motions.
 - a) Under damped harmonic motion
 - b) Over damped harmonic motion
 - c) Critically damped harmonic motion. Assume that the damping term is proportional to the velocity of the oscillation.
15. Write down the differences between free and forced oscillations. Give examples of each motions.
16. What are forced oscillations? Obtain an expression for the amplitude of the forced oscillator and give the condition for amplitude resonance.
17. Show that for a forced vibration, the total energy of the vibrating system is not constant.
18. Derive the expression of time period and frequency of oscillation for the spring-mass system.

19. Derive the expression of time period and frequency of oscillation for the spring-mass system when the mass of the spring is not equal to zero.
20. How does the energy consideration in a spring mass system affect the frequency of oscillation? Derive the mathematical expression of that frequency of oscillation.

Unit-2 Questions

1. A football is struck by lightning, causing it to acquire a uniform charge distribution. Using Gauss's Law, derive the expression for the electric field at a distance r from the center of the charged ball for the regions $r > R$ (outside the ball), $r = R$ (on the surface), and $r < R$ (inside the ball). Evaluate whether the electric field behaves differently in each region, and justify your reasoning based on electrostatic principles. Additionally, construct and interpret a graph of the electric field as a function of r for these three regions.
2. Derive equation of continuity in electromagnetics.
3. Define displacement current. Derive expression for Curl of magnetic field in terms of displacement current.
4. Using Gauss law, derive the expression for electric field due to a conducting and non-conducting sheet of charge.
5. Considering a charged conducting spherical shell and using Gauss's law of electrostatics, determine the electric field at various points. (a) inside the shell, (b) within the material of the shell, (c) outside the shell.
6. Write the the four Maxwell's equation in both integral and differential form. what was the correction made by Maxwell to the Ampere's law ?
7. What is displacement current, derive an expression for the same and what effect it has on Ampere's law?
8. What was the inconsistency in Ampere's law? how Maxwell modified the Ampere's law?
9. Given the symmetry of a long, uniformly charged wire, apply Gauss's Law of electrostatics to derive the expression for the electric field at a distance r from the wire.
10. Consider an infinite plane of charge with uniform surface charge density. Using Gauss's Law of electrostatics, derive the expression for the electric field at a point near the plane.
11. A piece of paper has acquired uniform charge density by rubbing. Determine the electric field at a distance of ' r ' from the paper, apply the Gauss's Law of electrostatics.
12. A fat wire of radius a , carries constant current I , uniformly distributed over its cross section. A narrow gap in the wire of width ($w < a$) form a parallel plate capacitor. Find the magnetic field in the gap at distance s from the axis ($s < a$)
13. Derive modified Amper's law in electrodynamics.
14. Derive equation of continuity in steady current.
15. Derive the differential form of Gauss law of electrostatics.
16. Calculate the electric field due to a non-conducting sheet of charge using Gauss law.
17. Derive expression for displacement current and explain its importance?
18. What happens to Maxwell's equations if there are no charges or current? will the electromagnetic field still exist, Justify your answer?
19. Derive differential form of Faraday's law of electromagnetic induction from integral form.

Unit-3 Questions

1. Newton's Rings: " i) Why the Central spot is dark, ii) Why circular rings are formed? iii) why alternate bright and dark fringes are formed.

2. List out of any 5 statements to identify, understand and characterise the wave nature of light waves.
3. Narrate the procedure to find the wavelength of the source using Newton's ring experimental set up.
4. Describe the intensity distribution in a Fraunhofer diffraction pattern produced by a single slit.
5. Explain the determination of wavelength of different colours using plane diffraction grating.
6. What is diffraction grating? Explain the formation of spectra when white light is diffracted by plane diffraction grating.
7. What is the output of natural light once passed through two crossed polarizers? What happens if a third polarizer is placed in between the the first two polarizers? You may use the law of Malus (Write malus law and one problem)
8. Narrate the procedure to find the radius of curvature of a plano-convex lens using a monochromatic light source with the help of Newton's ring experiment.
9. Illustrate the pictures to visualize the types of Interference: Division of wavefront and division of amplitude
10. Explain the differences between Fresnel and Fraunhofer classes of diffraction of light.
11. Describe the intensity distribution in the Fraunhofer diffraction pattern from a single slit, and explain why the central maximum is the brightest.
12. Write the grating equation and explain how a plane diffraction grating can be used to measure the wavelength of light.
13. What are the different types of polarized light?
14. A beam of polarized light with an initial intensity of $I_0 = 100 \text{ W/m}^2$ passes through two polarizers. The first polarizer is aligned with the initial polarization direction of the light, while the second polarizer makes an angle of 45 degrees with the first.
15. Do you know any methods to produce polarized light? explain
16. Give mathematical derivation for the production of linearly polarized light.
17. Give mathematical derivation for the production of circularly polarized light.
18. What is double refraction? explain.
19. Calculate the thickness of a quarter wave plate if the refractive indices of e-ray and o-ray are 1.543 and 1.499 respectively. Use wavelength of light as 5000 Å.

Unit-4 Sample questions

1	Does Heisenberg's Uncertainty principle applies to macroscopic particles?
2	What does the Heisenberg uncertainty principle imply?
3	State Heisenberg uncertainty principle.
4	What is the potential energy term for a particle outside the boundaries of a confined one-dimensional box?
5	Write the expression of the energy of a particle moving in a one dimensional infinite potential box.
6	Can the magnitude of a wave function ($\Psi^*(x,t) \times \Psi(x,t)$) be a negative number?
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8	If a photon and electron have same wavelength then which one has larger momentum?
9	The ratio of the de Broglie wavelengths of electron waves in two orbits is 3:5. Calculate the ratio of their kinetic energy?
10	What is the relationship between momentum (p) of the particle and wavelength (λ) of matter-wave.
11	If the momentum of a particle is increased by four times, what happen to its associated wavelength?
12	What do you mean by the orthogonality condition of the wavefunctions?
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14	$\Psi(x, t) = \psi(x)\exp(-iEt/\hbar)$ is a quantum state. Show that the probability density is independent of time.
15	What is the normalization condition?
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17	Write the relations of Heisenberg's Uncertainty principle interms energy, momentum and angle ?
18	If the uncertainty in the position of an electron is 4×10^{-10} m, calculate the uncertainty in its momentum and minimum uncertainty in its velocity.
19	Explain the Heisenberg Uncertainty Principle and its Implications for Measurement
20	Show that the solution of Schrodinger wave equation for particle in an infinite potential box leads to the concept of quantization of energy.
21	Find the probability of finding a particle trapped in a box of length 'L' in the region of 0.45L to 0.55L for its ground state.
22	A particle in a box exists in a superposition state: $\psi(x, t) = A (\psi_1 + \sqrt{3}\psi_2)$, where ψ_1 and ψ_2 are stationary states with quantum numbers $n=1$ and $n=2$ respectively. Calculate the normalization factor A?
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26	Calculate the de Broglie wavelength of an electron moving with one tenth of the velocity of light.
27	Given the wavefunction $\psi(x)=Ae^{\alpha x^2}$ where A and α are constants, what is the value of A required to normalize the wavefunction over all space?

Arun and Jay are playing tennis. Arun hits a tennis ball with a mass of 5 grams (0.005 kg), and the ball gains a velocity of 25 m/s.

(a) Calculate the de Broglie wavelength of the tennis ball.

(b) Based on the wavelength you calculate, determine whether this ball can be treated as a quantum particle. Justify your answer by discussing the conditions

28 under which quantum mechanical effects become significant.

29 What are matter waves and state deBroglie hypothesis.

30 Starting with time dependent Schrodinger's equation, obtain time independent Schrodinger's equation

31 Start from the plain wave, obtain to Schrodinger's equation.

32 Obtain the equation of continuity of the probability density, starting from the Schrodinger's equation.

33 Explain the physical significance of wave function.

34 Describe the properties of matter waves.

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A proton confined within a nucleus has a position uncertainty of 1×10^{-15} m. Estimate the minimum uncertainty in its momentum and hence its energy.

37 Given: $\hbar = 1.055 \times 10^{-34}$ J.s. proton mass $m_p = 1.67 \times 10^{-27}$ kg.

Starting from deBroglie wave concept, obtain Heisenberg's uncertainty principle?

38 Give one illustration of this principle.

Explain the Heisenberg Uncertainty Principle and its Implications for

39 Measurement

Write down the wavefunction for a free particle in a box of length L ($0 \leq x \leq L$).

Find out the normalization factor. Find the probability of finding the particle in the region $0 \leq x \leq L/2$.

40 Show that the solution of Schrodinger wave equation for particle in an infinite potential box leads to the concept of quantization of energy.

Write down the wave function for a quantum mechanical particle moving in a one dimensional infinite potential well of length L ($0 \leq x \leq L$). Calculate the expectation value of position (x) and linear momentum(p_x) for the ground state.

42 Reach to Time independent Schrodinger's equation from the time dependent Schrodinger's equation.

43 Start from the plain wave, reach to Schrodinger's equation.

Reach to the equation of continuity of the probability density, starting from the Schrodinger's equation.

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47 What are the limitations of Heisenberg's Uncertainty Principle?

48 What are the applications of Heisenberg's uncertainty principle?

49 State Heisenberg uncertainty principle.

Can the magnitude of a wave function ($\Psi^*(x,t) \times \Psi(x,t)$) be a negative number?

50 Explain.

Draw the energy levels for a free particle moving in an infinite potential box? Are

51 they equally spaced?

The wave function of a particle of mass m confined in one-dimensional infinite potential in the interval $0 \leq x \leq \pi$, is given by $\Psi(x) = N \sin(3x)$, where N is the normalization constant. Calculate N .

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The ratio of the de Broglie wavelengths of electron waves in two orbits is 3:5.

56 Calculate the ratio of their kinetic energy?

If a photon and electron have same wavelength then which one has larger

57 momentum?

What will happen to the wave function of a particle of mass m ,

58 when it hits the boundary of an infinite potential?

$\Psi(x, t) = \psi(x)\exp(-iEt/\hbar)$ is a quantum state.

59 Show that the probability density is independent of time.

60 What is the form of a plane wave? Can it be normalized?

Unit- 5 Sample questions

1	What is a LASER?
2	How is the population inversion is achieved in the lasing action?
3	What is the difference between normal light and laser light?
4	What is the role of chromium ions in the operation of a ruby laser.
5	What are the main components of a CO ₂ laser system
6	What is the significance of the optical cavity in a NdYAG laser?
7	Describe one application of laser speckle in biomedical imaging.
8	List two industrial applications of lasers and briefly explain their significance
9	Explain the principle behind the formation of laser speckles.
10	What is the significance of population inversion?
11	How can you calculate the ratio of spontaneous emission to stimulated emission?
12	List out five different uses of lasers in detail
13	Discuss the construction and working of a ruby laser with neat diagrams. Write any four applications of ruby laser
14	Describe the detailed construction and working principle of a CO ₂ laser with neat energy level diagram,
15	Explain the construction and working principle of the NdYAG laser with neat diagram, detailing the role of the active medium, pumping process, and optical cavity.
16	Distinguish between spontaneous and stimulated emission of radiation.
17	What is the process of stimulated emission?
18	Write about the properties of laser light.
19	Explain the working principle of a ruby laser, including the energy level transitions involved.
20	Explain the working principle of a Co2 laser, including the energy level transitions involved.
21	Discuss the operating principle of the Nd YAG laser, including the energy level transitions and the process of stimulated emission.
22	Describe the process of laser speckle formation in detail. How can speckle patterns be used to measure surface roughness?
23	Describe the characteristics of Laser Speckle.
24	Explain the formation of laser speckles and its applications.
25	What is the acronym of LASER?
26	What is the nature of spontaneous emission?
27	What is population inversion in case of laser light generation?
28	Which type of laser is the ruby laser?
29	In which region of the electromagnetic spectrum does the CO ₂ laser emit radiation?
30	What is the primary wavelength emitted by a Nd YAG laser?
31	What is a laser speckle pattern?
32	Name one common application of laser speckle imaging.
33	State one medical application of lasers.
34	what is basic Principles of Laser Speckle?
35	Write the working principle of laser.

36	What is the difference between lasers and non coherent sources?
37	Write down the in detail about a) absorption, b) stimulated emission c) stiumated absoprtion and d) spontaneous emission with diagram ?
38	Describe the detailed construction and working principle of a CO ₂ laser with neet energy level diagram, write any two applications of this laser
39	Discuss the construction and working of a ruby laser with neet diagrams. Write any four applications of ruby laser
40	Explain the construction and working principle of the NdYAG laser with neet diagram, detailing the role of the active medium, pumping process, and optical cavity.